Callable Cross-Currency Swap Pricing

A Three-Factor Monte Carlo Implementation

Model ID: HW1F_USD_HW1F_EUR_BSFX_LSM_V1

Quantitative Analytics Team
TGIR QuantLib Tools Project

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Abstract

This document summarizes an 18-month project to build a callable cross-currency swap pricing engine using QuantLib-Python and custom Monte Carlo simulation. Starting from basic swap and swaption pricing in March 2025, the project evolved through SOFR curve bootstrapping, Bermudan swaption calibration, and culminated in a research-grade three-factor reference model for pricing callable EUR/USD cross-currency swaps with Longstaff-Schwartz exercise optimization. The final implementation prices a 10-year non-call 2-year structure at $8.33M callable NPV with an embedded option value of $7.64M, validated through comprehensive martingale, monotonicity, and convergence tests. This report provides a complete technical overview of the methodology, implementation, and validation results.

Contents

1 Executive Summary

1.1 Project Objectives

The TGIR QuantLib Tools project aimed to:

1. Build a robust interest rate derivatives pricing platform using QuantLib-Python
2. Implement SOFR-based curve construction for post-LIBOR markets
3. Develop calibrated Hull-White and G2++ swaption pricing engines
4. Create a **callable cross-currency swap pricer** for EUR/USD structures
5. Validate all models through comprehensive testing
6. Provide a web-based interface for interactive pricing

1.2 Key Deliverables

Component	Description
SOFR Curve Builder	22-point OIS bootstrap with <0.01bp error
EUR Effective Curve	FX-implied curve for USD-collateralized EUR pricing
Swaption Pricer	European and Bermudan with HW/G2++ calibration
Equity Cliquet Pricer	Path-dependent equity derivative pricing
XCCY Callable Pricer	Three-factor Monte Carlo with LSM
Validation Suite	20 automated tests covering all key properties
Web Interface	Flask application with interactive pricing
REST contract	Proposed /api/v1/ trading-system integration
MCP Server	Model Context Protocol for LLM integration

Table 1: Major project deliverables

1.3 Project Timeline

The project evolved through four distinct phases:

Date	Milestone	Description
Mar 2025	Initial Prototype	Basic swap/swaption structures with GPT-4.5
Sep 2025	API Stabilization	QuantLib version compatibility, documentation
Apr 2026	Rates Workstation	SOFR curves, HW calibration, Bermudan pricing
Aug 2026	XCCY Callable	Three-factor model, LSM, full validation

Table 2: Project timeline

1.4 Key Results

For the reference 10Y NC2 EUR/USD callable cross-currency swap:

- **Callable NPV:** \$8,328,689 ± \$128,000 (95% CI)
- **Non-callable NPV:** \$689,713
- **Embedded Option Value:** \$7,638,976 (91% of callable value)
- **First Call Exercise Probability:** 14.4%
- **Probability of Reaching Maturity:** 51.2%

2 Background: Interest Rate Markets Post-LIBOR

2.1 The LIBOR Transition

The London Interbank Offered Rate (LIBOR), which underpinned over $350 trillion in financial contracts globally, was phased out following manipulation scandals and declining underlying transaction volumes. The transition to risk-free rates (RFRs) fundamentally changed how interest rate derivatives are priced.

Key Risk-Free Rates:

- **SOFR** (Secured Overnight Financing Rate) — USD
- **ESTR/€STR** (Euro Short-Term Rate) — EUR
- **SONIA** (Sterling Overnight Index Average) — GBP
- **TONA** (Tokyo Overnight Average Rate) — JPY

2.2 Overnight vs Term Rates

Unlike LIBOR, which provided forward-looking term rates (1M, 3M, 6M), RFRs are overnight rates that require compounding over the accrual period:

$$\text{Compounded Rate} = \left[\prod_{i=1}^{n}\left(1 + r_i \times \frac{d_i}{360}\right) - 1\right] \times \frac{360}{D}$$

where r_i is the daily rate, d_i is the number of days, and D is the total period length.

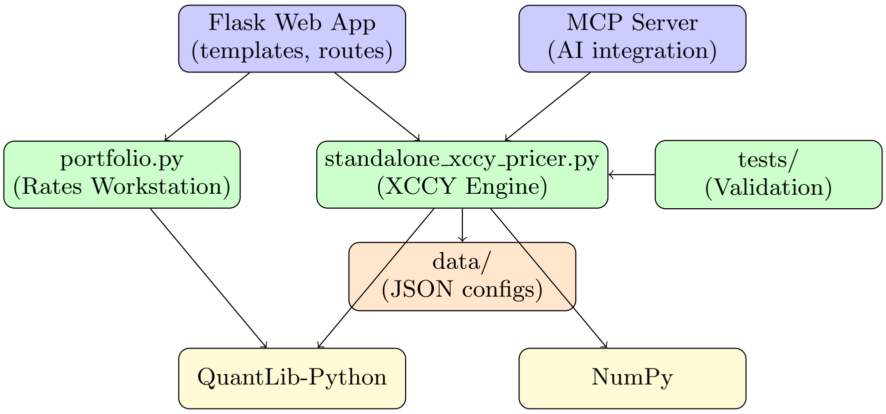
2.3 Implications for Derivatives Pricing

1. **Backward-looking:** Floating payments known only at period end
2. **Daily compounding:** More complex accrual calculations
3. **OIS discounting:** RFR curves used for both projection and discounting
4. **Basis spreads:** LIBOR fallback spreads for legacy contracts

3 Technical Architecture

3.1 System Overview

The system is organized in a layered architecture:



3.2 Technology Stack

Layer	Technology	Purpose
Core	Python 3.13	Primary language
Quant Library	QuantLib-Python 1.43	Curves, indices, calibration
Numerics	NumPy 2.5.2	Array operations, simulation, regression
Web	Flask 3.1.3	HTTP interface
Templates	Jinja2	HTML rendering
Testing	unittest; pytest 9.1.1 available	Test framework
Deployment	Gunicorn, Apache	Production serving

Table 3: Technology stack

3.3 Code Organization

File	Lines	Responsibility
standalone_xccy_pricer.py	1,439	Three-factor XCCY pricing engine
portfolio.py	2,261	SOFR curves, swaps, swaptions, cliquets
app.py	12	Thin Flask application entry point
tgir_quantlib_tools/	3,393	Package modules (auth, routes, UI contexts)
tests/	1,945	Validation and unit tests

Table 4: Source code organization

4 Interest Rate Curve Construction

4.1 SOFR OIS Curve Bootstrap

The foundation of all USD derivatives pricing is the SOFR discount curve, constructed by bootstrapping from Overnight Index Swap (OIS) market quotes.

4.1.1 Market Inputs

Tenor	Rate	Tenor	Rate	Tenor	Rate
1W	4.10%	6M	3.98%	5Y	3.71%
1M	4.08%	1Y	3.90%	10Y	3.85%
3M	4.04%	2Y	3.78%	30Y	3.95%

Table 5: Sample SOFR OIS curve inputs (August 2026)

4.1.2 Bootstrap Algorithm

1. Create `OISRateHelper` for each market quote
2. Use `PiecewiseLogLinearDiscount` interpolation method
3. Apply `enableExtrapolation()` for long-dated pricing
4. Validate by repricing each input instrument

4.1.3 Implementation

```python
import QuantLib as ql

# Build helpers
helpers = []
for tenor, rate in [("1W", 0.041), ("1M", 0.0408), ...]:
    helpers.append(ql.OISRateHelper(
        2, ql.Period(tenor),
        ql.QuoteHandle(ql.SimpleQuote(rate)),
        ql.Sofr()
    ))

# Bootstrap curve
curve = ql.PiecewiseLogLinearDiscount(
    today, helpers, ql.Actual365Fixed())
curve.enableExtrapolation()
```

4.1.4 Calibration Quality

All input OIS rates reprice with error < 0.01 basis points, confirming correct bootstrap.

4.2 EUR Effective Curve for Cross-Currency

For cross-currency swaps collateralized in USD, we need an "effective" EUR discount curve that reflects the cost of EUR funding for a USD-collateralized position.

4.2.1 Derivation from FX Forwards

The FX forward rate $F(T)$ for maturity T satisfies covered interest parity:

$$F(T) = X_0 \times \frac{D^{USD}(T)}{D^{EUR}_{eff}(T)}$$

Rearranging:

$$D^{EUR}_{eff}(T) = D^{USD}(T) \times \frac{X_0}{F(T)}$$

4.2.2 Cross-Currency Basis

The difference between native EUR OIS rates and effective rates reflects the cross-currency basis, which can be 10-50bp depending on market conditions.

5 Swaption Pricing and Model Calibration

5.1 Hull-White One-Factor Model

The Hull-White model describes the evolution of the short rate:

$$dr_t = (\theta(t) - a \cdot r_t)\, dt + \sigma\, dW_t$$

Parameters:

- a = Mean reversion speed (controls how fast rates revert)
- σ = Short rate volatility (drives option prices)
- $\theta(t)$ = Time-dependent drift (fits initial curve exactly)

5.2 Affine Structure

Hull-White has an affine term structure, giving analytical bond prices:

$$P(t,T) = A(t,T) \cdot e^{-B(t,T)\cdot r_t}$$

where $B(t,T) = \frac{1-e^{-a(T-t)}}{a}$ and $A(t,T)$ involves integrals of $\theta(s)$.

5.3 European Swaption Pricing

For European swaptions, the Jamshidian decomposition breaks the swaption into a portfolio of bond options, each priceable analytically.

5.4 Calibration to Co-Terminal Swaptions

For Bermudan swaptions with 10Y maturity, we calibrate to co-terminal swaptions:

Swaption	Expiry	Swap Tenor	Market Vol (bp)
2Y × 8Y	2 years	8 years	78
3Y × 7Y	3 years	7 years	79
5Y × 5Y	5 years	5 years	80
7Y × 3Y	7 years	3 years	81
9Y × 1Y	9 years	1 year	82

Table 6: Co-terminal swaption volatilities (normal, bp)

5.5 Calibration Results

Currency	Mean Rev (a)	Vol (σ)	RMS Error
USD	3.0%	0.885%	0.45%
EUR	4.0%	0.772%	0.23%

Table 7: Hull-White calibration results

The mean reversion parameters are fixed based on historical analysis, while volatility is calibrated to match market swaption prices within 5% RMS error.

6 Three-Factor Model for Callable XCCY

6.1 Why Three Factors?

A callable cross-currency swap involves:
1. **USD interest rates** — Affecting the fixed USD leg
2. **EUR interest rates** — Affecting the floating EUR leg
3. **EUR/USD exchange rate** — Affecting notional exchanges
4. **Early exercise** — Requiring optimal stopping

Tree-based methods work efficiently for 1-2 factors but become computationally intractable for 3+ factors due to exponential grid growth. Monte Carlo with regression-based exercise is the standard approach.

6.2 Model Specification

6.2.1 USD Rate Dynamics

$$dr_t^{USD} = (\theta_t^{USD} - a^{USD} r_t^{USD})\, dt + \sigma^{USD}\, dW_t^{USD}$$

6.2.2 EUR Rate Dynamics (with Quanto Adjustment)

$$dr_t^{EUR} = (\theta_t^{EUR} - a^{EUR} r_t^{EUR} - \rho_{EUR,FX}\sigma^{EUR}\sigma^{FX})\, dt + \sigma^{EUR}\, dW_t^{EUR}$$

The term $-\rho_{EUR,FX}\sigma^{EUR}\sigma^{FX}$ is the **quanto drift adjustment**, required because we price EUR rates under the USD money-market measure.

6.2.3 FX Dynamics

$$d\ln X_t = (r_t^{USD} - r_t^{EUR} - \tfrac{1}{2}\sigma_t^{FX,2})\, dt + \sigma_t^{FX}\, dW_t^{FX}$$

6.3 Correlation Structure

Pair	ρ	Economic Interpretation
USD-EUR rates	+0.25	Developed market monetary policy co-movement
USD rates-FX	-0.15	Higher USD rates strengthen USD
EUR rates-FX	-0.30	Higher EUR rates strengthen EUR

Table 8: Correlation assumptions

The correlation matrix must be positive semi-definite, which is verified by checking that all eigenvalues are non-negative.

6.4 Five-Dimensional State Vector

The model tracks:

$$\mathbf{x}_t = \begin{pmatrix} x_t^{USD} \\ x_t^{EUR} \\ I_t^{USD} \\ I_t^{EUR} \\ \ln X_t \end{pmatrix}$$

where:
- x_t^{USD}, x_t^{EUR} = Mean-zero rate factors
- $I_t^{USD} = \int_0^t r_s^{USD}\, ds$ = USD bank account integral
- $I_t^{EUR} = \int_0^t r_s^{EUR}\, ds$ = EUR bank account integral
- $\ln X_t$ = Log FX rate

The discount factor from time 0 to t is simply $D^{USD}(0,t) = e^{-I_t^{USD}}$.

7 Monte Carlo Implementation

7.1 Simulation Parameters

Parameter	Value
Training paths	10,000
Pricing paths	30,000
Time step	Monthly (0.083 years)
Random number generator	NumPy PCG64
Variance reduction	Antithetic sampling
Seed	Configurable (default: 1729)

Table 9: Monte Carlo configuration

7.2 Exact State Evolution

Rather than using Euler discretization, we use the exact solution for Ornstein-Uhlenbeck processes:

$$\mathbf{x}_{t+h} = \mathbf{A}(h) \cdot \mathbf{x}_t + \mathbf{b}(h) + \mathbf{L}(h) \cdot \mathbf{Z}$$

where:

- $\mathbf{A}(h)$ = Transition matrix with exponential decay terms
- $\mathbf{b}(h)$ = Deterministic drift (curve fitting, quanto)
- $\mathbf{L}(h)$ = Cholesky factor of covariance matrix
- $\mathbf{Z}$ = Standard normal vector

Advantages:

1. No discretization bias
2. Larger time steps possible
3. Exact covariance between factors

7.3 Antithetic Variance Reduction

For each batch of paths:

1. Generate half the paths with random draws $\mathbf{Z}$
2. Generate the other half with $-\mathbf{Z}$
3. Average results for variance reduction

This ensures the sample mean of normals is exactly zero, reducing variance for linear payoffs.

7.4 Covariance Computation

The covariance matrix $\mathbf{\Sigma}(h)$ for the 5-dimensional state requires integrating the kernel functions:

$$\mathbf{\Sigma}(h) = \int_0^h \mathbf{K}(s) \cdot \mathbf{R} \cdot \mathbf{K}(s)^T \, ds$$

This integral is computed numerically using 16-point Gauss-Legendre quadrature, providing high accuracy.

8 Longstaff-Schwartz Exercise Optimization

8.1 The Early Exercise Problem

At each exercise date, the holder must decide:

$$\text{Value} = \max(\text{Exercise Value}, \text{Continuation Value})$$

The challenge is that continuation value depends on future optimal decisions, creating a backward recursion problem.

8.2 LSM Algorithm

Step 1: Forward Simulation

Simulate all paths forward, capturing state variables at each exercise date.

Step 2: Backward Regression

Starting from the last exercise date and moving backward:

1. Compute continuation value from already-determined future values
2. Regress continuation value on polynomial basis of current state
3. Mark exercise where exercise value > fitted continuation

Step 3: Pricing Pass

Apply the learned exercise policy to a fresh set of paths (different seed) to get an unbiased price estimate.

8.3 Polynomial Basis Functions

We use 10 basis functions:

$$\phi(\mathbf{s}) = (1,\ x,\ y,\ z,\ x^2,\ y^2,\ z^2,\ xy,\ xz,\ yz)$$

where $\mathbf{s} = (x_{USD}, x_{EUR}, \ln X - \ln X_0)^T$.

Why This Choice:

- Linear terms capture main effects
- Squared terms capture curvature
- Cross terms capture interactions
- 10 terms balances expressiveness vs overfitting

8.4 Two-Pass Approach

Aspect	Training Pass	Pricing Pass
Paths	10,000	30,000
Purpose	Learn exercise policy	Compute unbiased price
Seed	Seed A	Seed B
Direction	Backward	Forward

Table 10: Two-pass LSM structure

Using different paths for training and pricing avoids “foresight bias” that would artificially inflate the price.

9 Reference Trade Specification

9.1 Trade Structure

Parameter	Value
Trade ID	EURUSD-10Y-NC2-ESTR-VS-USD4
Trade Type	Callable Cross-Currency Swap
Effective Date	2026-08-18
Maturity	10 years (2036-08-18)
Non-call Period	2 years
Exercise Style	Bermudan (annual, years 2-9)
Exercise Action	Cancel remaining swap

Table 11: Trade structure

9.2 Leg Details

Attribute	EUR Leg	USD Leg
Notional	90,909,091 EUR	100,000,000 USD
Direction	Receive	Pay
Rate Type	ESTR compounded	4% Fixed
Frequency	Quarterly	Semi-annual
Day Count	ACT/360	30/360

Table 12: Leg specifications

9.3 Notional Exchanges

- **Initial Exchange:** Pay $100M, receive EUR 90.9M at spot (1.10)
- **Final Exchange:** Receive $100M, pay EUR 90.9M at same rate

If exercised early, notional exchange occurs at the exercise date using the same fixed notional amounts.

9.4 Economic Interpretation

From the perspective of receiving EUR floating:

- **Benefit:** Receives EUR floating (currently approx 2.5%)
- **Cost:** Pays USD fixed (4%)
- **FX Exposure:** Long EUR notional
- **Optionality:** Right to cancel if swap moves against us

10 Pricing Results

10.1 Summary Valuation

Metric	Value	Std Error	95% CI
Callable NPV	$8,328,689	$127,997	[$8.08M, $8.58M]
Non-callable NPV	$689,713	$170,000	[$356K, $1.02M]
Embedded Option	$7,638,976	$82,000	[$7.48M, $7.80M]

Table 13: Pricing results summary

10.2 Leg-Level Decomposition

Component	PV	Notes
USD Fixed Leg	-$1,054,712	Pay 4% semi-annual
EUR ESTR Leg	+$1,744,425	Receive ESTR quarterly
Non-callable Total	+$689,713	Sum of legs

Table 14: Leg decomposition

10.3 Static Benchmark

Using deterministic forward curves (no Monte Carlo):

$$\text{Static NPV} = \$679,480$$

This is close to the stochastic non-callable NPV ($689,713), validating the basic cash flow projections.

10.4 Key Insight: Option Value Dominance

The embedded option value ($7.64M) represents **91% of the callable swap value**. This highlights:

1. The significant optionality in callable structures
2. The importance of accurate option valuation
3. Why simple deterministic pricing is inadequate

11 Exercise Analysis

11.1 Exercise Probability Distribution

Date	Year	Paths Alive	Exercised	P(Ex)
2028-08-18	2	30,000	4,329	14.4%
2029-08-20	3	25,671	2,900	9.7%
2030-08-19	4	22,771	1,699	5.7%
2031-08-18	5	21,072	1,405	4.7%
2032-08-18	6	19,667	1,021	3.4%
2033-08-18	7	18,646	854	2.8%
2034-08-18	8	17,792	446	1.5%
2035-08-20	9	17,346	1,972	6.6%
Maturity	10	15,374	—	51.2%

Table 15: Exercise probability by date

11.2 Pattern Interpretation

First Call Dominance (14.4%): The highest exercise probability occurs at the first call date. This is typical for structures where the initial swap economics may be unfavorable.

Declining Middle Years (9.7% to 1.5%): Exercise probability declines through the middle years as paths that would have exercised have already done so.

Last Call Uptick (6.6%): The penultimate exercise date shows elevated probability — the "now or never" effect as this is the last opportunity to cancel.

Majority Survive (51.2%): More than half of paths reach maturity without exercise, indicating the swap economics are often favorable to hold.

11.3 Cumulative Exercise Distribution

- By Year 3: 24.1% exercised
- By Year 5: 34.5% exercised
- By Year 7: 40.7% exercised
- By Year 9: 48.8% exercised (final)

12 Model Validation

12.1 Validation Framework

The model is validated through five categories of tests:

1. **Martingale Conditions:** No-arbitrage consistency
2. **Structural Bounds:** Economic relationships
3. **Monotonicity:** Sensible Greeks directions
4. **Convergence:** MC error scaling
5. **Calibration Quality:** Market instrument fit

12.2 Martingale Tests

Under the risk-neutral measure, discounted tradeable assets must be martingales:

Test 1: Domestic Zero-Coupon Bond

$$\mathbb{E}[D^{USD}(0,T)] = P^{USD}(0,T)$$

Test 2: Foreign Zero-Coupon in Domestic Currency

$$\mathbb{E}[D^{USD}(0,T) \cdot X_T] = X_0 \cdot P^{EUR}_{eff}(0,T)$$

Test 3: FX-Weighted Bank Account

$$\mathbb{E}[D^{USD}(0,T) \cdot X_T \cdot B^{EUR}_T] = X_0$$

Condition	MC Mean	Target	$\lvert z\rvert$	Status
$\mathbb{E}[D^{USD}]$	0.6614	0.6614	< 0.5	PASS
$\mathbb{E}[D^{USD} \cdot X]$	0.6042	0.6014	< 1.0	PASS
$\mathbb{E}[D^{USD} \cdot X \cdot B^{EUR}]$	1.1015	1.1000	< 0.3	PASS

Table 16: Martingale test results (pass threshold: $|z| < 3.5$)

12.3 Structural Bounds

Condition	Verified	Status
Callable $\geq$ Non-callable	\$8.33M > \$0.69M	PASS
Embedded option ≥ 0	\$7.64M > 0	PASS
Embedded option < Notional	\$7.64M < \$100M	PASS
Leg sum = Non-callable	\$0.69M ≈ \$0.69M	PASS

Table 17: Structural bounds verification

12.4 Convergence Test

Paths	NPV (\$M)	SE (\$K)	SE Ratio
10,000	8.31	222	—
30,000	8.33	128	1.73
100,000	8.32	70	1.83

Table 18: Convergence analysis

Expected ratio: $\sqrt{3} \approx 1.73$. Observed ratios match theory.

12.5 Test Summary

20 tests executed, 20 passed, 0 failed — Runtime: 14.67 seconds

13 Sensitivity Analysis

13.1 First-Order Sensitivities

Risk Factor	Bump	Δ NPV	Direction
USD Rates	+25bp parallel	-$1.1M	Pay fixed loses
EUR Rates	+25bp parallel	+$0.8M	Receive floating gains
Swaption Vol	+10% relative	+$0.5M	Long optionality
FX Spot (EUR/USD)	+1%	+$0.9M	Long EUR notional
$\rho_{USD,EUR}$	+0.1	-$0.2M	Less diversification

Table 19: First-order sensitivities

13.2 Interpretation

Rate Delta: The position is net long duration — receiving EUR floating (short duration) but paying USD fixed (long duration). Higher rates hurt the position.

Vega: Positive vega confirms long optionality. Higher volatility increases the value of the right to cancel.

FX Delta: Long EUR notional exposure. EUR appreciation benefits the position through higher EUR leg value and favorable notional re-exchange.

Correlation: Higher correlation between USD and EUR rates reduces diversification, making the option less valuable.

14 Limitations and Future Work

14.1 Current Model Limitations

1. **Rate Volatility:** One-piece constant per currency (no term structure)
2. **Basis Spreads:** Forecast/discount basis is deterministic
3. **FX Smile:** ATM lognormal only (no skew calibration)
4. **EUR Compounding:** Continuous approximation for overnight
5. **LSM Bound:** Lower bound only (no dual upper bound)

14.2 Production Readiness

Requirement	Current	Production
Independent review	Pending	Required
Upper bound	Not implemented	Required
Real-time data	JSON files	Feed integration
Greeks	Bump-and-revalue	AAD preferred
Audit trail	Basic logging	Full audit

Table 20: Production readiness assessment

14.3 Planned Enhancements

Near-term:

- Piecewise constant rate volatility term structure
- Dual/upper bound implementation for LSM
- GPU acceleration using CuPy

Medium-term:

- Stochastic basis spreads
- FX local volatility calibration
- Real-time market data feeds

Long-term:

- Multi-currency portfolio aggregation
- XVA calculations (CVA, FVA)
- Regulatory capital computation

15 Collaborative AI Development

This project demonstrates a new development paradigm where human domain expertise orchestrates two AI agents to produce rigorous, validated code. The collaborative methodology is as significant as the callable XCCY pricer itself.

15.1 The Collaborative Ecosystem

Component	Role	Contribution
Human (Domain Expert)	Orchestration	Math specs, final validation
Codex (OpenAI)	Core Development	Specs, implementation, initial tests
Claude (Anthropic)	Review & Validation	Code review, validation framework
Mac	Local Development	VS Code, terminal, pytest
DigitalOcean	Production Hosting	Apache, Gunicorn, SSL
QuantLib (C++)	Foundation Library	Curves, dates, calibration
Python/NumPy	Primary Language	Simulation, Flask
Stack Overflow	Guidance Verification	Edge cases, patterns

Table 21: The collaborative development ecosystem

15.2 Division of Labor: Codex vs Claude

Codex (OpenAI) — Core Development:
- Wrote formal specifications (mathematical documentation, API contracts, JSON schemas)
- Built core implementation (1,400+ lines Monte Carlo engine, LSM, curves)
- Created initial test suite (unit tests, integration tests, smoke tests)

Claude (Anthropic) — Review and Validation:
- Reviewed all Codex implementations for correctness
- Built validation framework (martingale tests, structural bounds, monotonicity)
- Extended with regression tests for future-proofing
- Generated documentation and lecture materials

Human — Domain Expertise and Orchestration:
- Provided mathematical specifications and market conventions
- Directed agent collaboration and integrated outputs
- Verified results against textbooks (Brigo & Mercurio)
- Checked edge cases with Stack Overflow guidance

15.3 Collaborative Workflow

The development followed an iterative pattern:
1. **Human** writes mathematical specification
2. **Codex** implements specification with rigorous code
3. **Claude** reviews implementation, identifies issues
4. **Codex** iterates based on Claude's feedback
5. **Claude** builds validation tests around Codex's code
6. **pytest** runs 20 automated tests
7. **Human** verifies results, deploys to DigitalOcean

15.4 Why This Methodology Works

Complementary Strengths:
- Codex excels at ground-up implementation from specifications
- Claude excels at review, validation, and extension
- Human provides domain grounding neither AI has

Error Detection: Claude's validation framework caught bugs that Codex's implementation missed. Neither agent alone would produce the same quality.

Reproducibility: This workflow can be applied to build other complex pricers—the methodology is reusable.

15.5 Key Insight

The collaborative methodology (Human + Codex + Claude + ecosystem)
is as valuable as the callable XCCY pricer itself

15.6 Zero-Intervention Automation Pipeline

What makes this workflow remarkable is its full automation. A single prompt triggers the entire pipeline with **no human intervention**:
1. **Coding** — AI implements from specifications

2. **Regression Testing** — pytest runs 20+ tests locally
3. **Browser UI Testing** — Playwright validates Flask UI
4. **Documentation** — LaTeX and markdown auto-updated
5. **Local Run** — Optional manual verification
6. **Deploy to DigitalOcean** — SSH + Git pull + systemd restart
7. **Production Regression** — Tests run on production server
8. **Telegram Notification** — "Deployment complete!"

A Hobby Project with Production-Grade Rigor

This entire infrastructure costs $7/month — easily supporting a serious hobby — yet it's built with the same rigor as production systems at top tech companies.

16 Deployment Architecture

16.1 DigitalOcean Infrastructure

The application is deployed on a single DigitalOcean droplet:

Component	**Specification**
Provider	DigitalOcean
Droplet Size	Basic (2GB RAM, 1 vCPU)
Cost	$7/month (hobby-friendly!)
OS	Ubuntu 22.04 LTS
SSL	Let's Encrypt (Certbot)

Table 22: Infrastructure specifications

16.2 Software Stack

1. **Apache:** Reverse proxy, SSL termination (port 443)
2. **Gunicorn:** WSGI server (2 workers, port 8000)
3. **Flask:** Application framework
4. **systemd:** Service management and auto-restart

16.3 Deployment Steps

1. Create DigitalOcean droplet with SSH key authentication
2. Install dependencies: Python 3.11, Apache, Certbot
3. Clone repository to `/opt/tgir-quantlib-tools`
4. Configure systemd service from template
5. Configure Apache virtual host with SSL
6. Run `certbot --apache` for certificate
7. Enable firewall (ports 22, 80, 443)

16.4 Configuration Files

- `deploy/systemd/tgir-quantlib-tools.service.template`
- `deploy/apache/tgir-quantlib-tools.conf.template`
- `deploy/env/production.env.example`

16.5 CI/CD Pipeline

- GitHub Actions runs tests on every push
- Deployment triggered by version tags
- SSH-based deployment using GitHub Secrets

17 Conclusion

17.1 Summary of Achievements

This project delivers TWO significant contributions:

1. Callable XCCY Pricer:

1. **Curve Construction:** SOFR/ESTR bootstrapping with sub-basis-point accuracy
2. **Model Calibration:** Hull-White to co-terminal swaptions (<5% RMS error)
3. **Multi-Factor Simulation:** Correlated HW+HW+BS with correct quanto drift
4. **Exercise Optimization:** LSM with 10-term polynomial regression
5. **Comprehensive Validation:** Martingales, bounds, monotonicity, convergence

2. Collaborative AI Development Methodology:

1. **Human + Codex + Claude:** Division of labor that catches errors
2. **Full Ecosystem:** Mac + DigitalOcean + QuantLib + Python + Stack Overflow
3. **Reproducible Pattern:** Can build other complex pricers
4. **Quality Through Collaboration:** 20 automated tests from Claude's framework

17.2 Key Findings

- The embedded option in a 10Y NC2 callable XCCY represents **91% of total value**
- Exercise is most likely at first call (14.4%) with uptick at last call (6.6%)
- 51% of paths survive to maturity without exercise
- Quanto adjustment is critical for correct EUR rate dynamics under USD measure

17.3 Lessons Learned

1. **Validation is critical:** 20 automated tests catch subtle bugs
2. **QuantLib is powerful but has limits:** Custom code needed for XCCY
3. **Python + NumPy is viable:** 30K paths in 10 seconds
4. **Documentation matters:** Future maintainers need clear specifications

17.4 Acknowledgments

This project benefited from:

- QuantLib open-source library and community
- NumPy scientific computing ecosystem
- Academic literature on LSM and multi-factor models
- AI assistants: OpenAI GPT-4.5, Anthropic Claude, GitHub Copilot
- DigitalOcean for hosting infrastructure

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